

PK ENTITY LINKING ANNOTATION GUIDELINES

Project Background

Biomedical Literature contains a wealth of information on Pharmacokinetics (PK), including parameters and covariates, often reported in free-text data. This data is helpful for preclinical PK predictions and initiating Population PK Models. However, it takes time for PK researchers to extract information from the literature manually. Extracting this data automatically from free text would enable accelerated curation of PK databases for research and drug development.

Computationally, automatically extracting data from sentences is a complex task. The machine needs to (a) recognise the entities within sentences (e.g., PK parameters), known as named entity recognition (NER), (b) be able to standardise these entities by linking different surface forms to a single form in a structured knowledge base or ontology, known as entity linking, and (c) understand the relations between entities in sentences, known as relation extraction (RE).

Aim: In this study, we aim to develop an automated entity-linking approach to link the PK parameter mentions in sentences to concepts in a PK ontology, to complement existing NER and RE approaches. This addresses step (b) of automated data extraction from free text in PK.

Method: We sample sentences from abstracts of PubMed papers reporting PK parameters. Then, we ask annotators to label these PK parameters as concepts from the PK Parameter ontology.

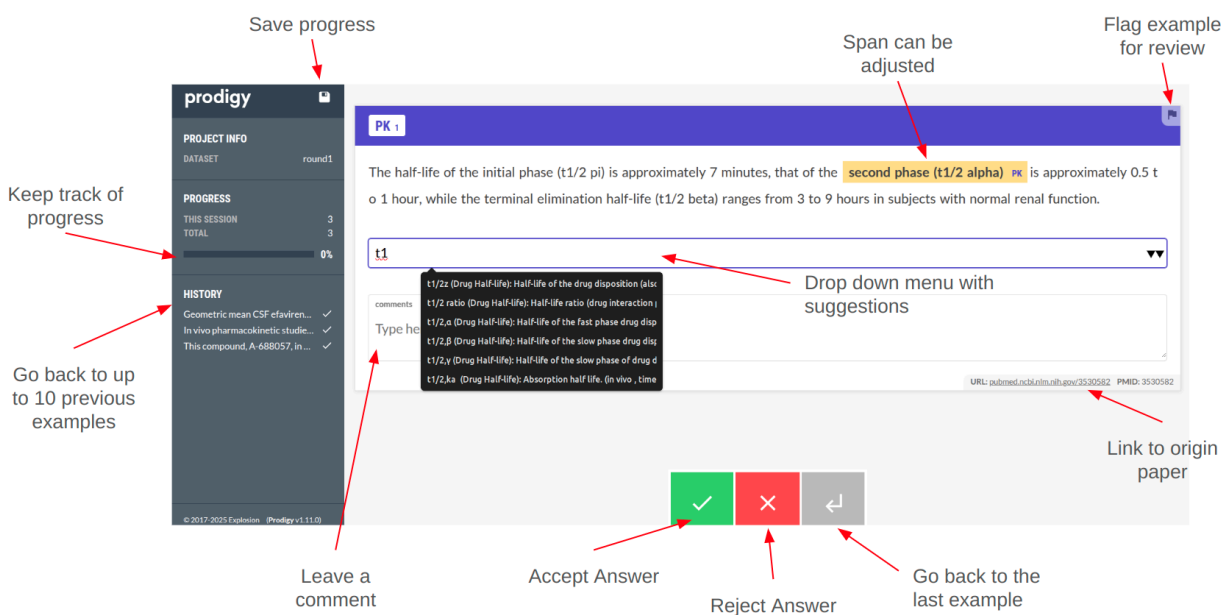
Task Overview & Annotation Interface

The annotation interface displays a single sentence with a single highlighted mention of a PK parameter. It also shows a list of selectable options representing the concepts in the PK parameter ontology, including a “NIL” option for irrelevant (e.g. not a PK parameter in the ontology). The annotator must select a label from these possible choices for the mention. The annotator should ignore any other PK parameter mentioned in the text shown and only focus on the highlighted span. Two annotators are assigned to annotate each table; a third annotator checks disagreements and raises these to discuss with the team where necessary.

The main subtasks are as follows:

1. First, Check if the initial PK entity span at the top is correct; you can re-highlight it if there is an error.
2. Secondly, choose the PK entity subtype from the list. Use the “NIL” label to mention things not associated with an entity in the list.
3. Please note that the interface will not let you proceed to the following example if you do not choose an option from the drop-down list. Therefore, if unsure, please select the closest option or “NIL”, flag the example and leave a comment.

Details of the key functionality of the labelling interface are shown below:



Saving Annotations: If your annotations are saved correctly, you will see a green box in the right-hand corner when you press the floppy disk (in the top left-hand corner). If this box is red, an error occurs on the server, so please raise this immediately.

Viewing Guidelines: You can quickly view these guidelines inside the interface by pressing the question mark in the top left-hand corner.

Further Information: A link to the original paper abstract from which the sentence is taken is provided to allow information to be checked against the paper text if needed.

What if I'm unsure? Please flag the example using the flag icon in the top right-hand corner and leave a comment in the box. The team can then review and discuss it.

Summary of Doubt/Clarifications

Dealing with NER model span errors:

- **Incorrect Labels e.g. non-PK Parameters:** If the highlighted span refers to any of the following, please label them as “NIL”:
 - Part of a PK/PD parameter is highlighted (e.g. AUC/MIC).
 - A pure PD parameter is highlighted (e.g., MIC, Emax, Maximum tolerated dose (MTD), or time to disease progression).
 - Any other text that does not refer to in vivo PK parameters, e.g., physiological parameters, chemical parameters, or clinical endpoints.
- **Incorrect Spans:** If the highlighted span refers to a PK parameter, but the span is not correct, please correct the spans, for example:
 - If only part of the full span has been highlighted (e.g., if the model highlighted “Cl” of the full term “Clearance”), then please correct this to highlight the full span.
 - If the parameter highlighted includes extra text at the beginning or end of the span that is not part of the parameter itself (e.g., “maximum concentration of baclofen”), please shorten the span (e.g., to “maximum concentration” only).
 - However, if a composite entity refers to a specialised version of a PK parameter (e.g. “area under the midazolam concentration curve”, “AUC_plasma”) then link the entire span to the core PK parameter in the ontology.
 - In cases such as variability on parameters (e.g. IIV Cl) or parameter covariate effects (e.g. weight on Cl), only link the parameter mention itself (e.g. Cl) to the ontology.
 - Please note if the highlighted span does not refer to a PK parameter, please **do not** adjust/correct it, just label as “NIL”.

Hints and Tips:

- If a PK parameter is highlighted but is not included in our ontology, please include it but mark it with the “NIL” label.
- In cases where the stated parameter is similar to our labels or a combination of two or more of our labels, but not one specific type, we will mark it as NIL, for example:

- “Cmax(ss)” combines C_{ss} and C_{max} from our ontology. We don’t have this combined parameter as an option, so link to NIL.
 - “t_{1/2} pi” or “t_{1/2}n” refers to very fast distribution phase before t_{1/2}alpha. - This is not included in our ontology, so it is classified as NIL.
- Cumulative excretion or urinary recovery rates are marked as “Q44:fe” if the units are % or a fraction. If the units are in mass, mark them as “Q91:Ae”, e.g.

PK 1

The **urinary recovery rates** PK were determined in the first 8 hours after administration in the 5 mg/kg and 6 hours in the 10 mg/kg group, and the values were as low as 1.05-12.3% and 1.6-4.33%, respectively.

comments

In mass balance studies the urine recovery of radio-labelled isotopes are quite important to understand the importance of renal and bile elimination: urine radio isotope recovery (%).

- “Plasma clearance” after oral administration refers to “Q27:CL/F”; if it is after IV administration, label it “Q22:CL”.
- When encountering a composite entity referring to multiple entities in our ontology, choose the label from the ontology which matches the nearest numeric value e.g. for “the half-lives were x, y and z for phases in a three-compartment model”, choose the parameter which matches x.

PK 1

Although leuprolide disappeared rapidly from the serum after intravenous administration (the **biological half-lives** PK were 8.4 min in the alpha-phase and 33.2 min in the beta-phase), long-lasting serum levels were observed when the analogue was administered vaginally.

comments

Mean both alpha and beta half-lives.

- Careful with wording - In this case, it initially looks like “Q36:Cmin” but is talking about a dose to give to the patients, so it is “Q100:NIL”.

PK 1

In the treatment of patients with refractory schizophrenia, a **minimum concentration** PK of 350 micrograms/L was suggested as needed.

- Oral bioavailability is marked as absolute bioavailability (“Fab”).

PK 1

The intestinal, hepatic, and **oral bioavailability** PK of fructose was similar between 45% glucose/55% fructose and sucrose.

- Look out for the TBC, which can stand for Total Body Clearance.

PK 1

The corresponding mean values for group II are Cp3h 3305 ng/ml, AUC 15080 ng x h/ml, **TBC** PK 20.48 l/h/m², RC 7.02 l/h/m² and 26.23%.

- $AUC(0-\tau)$ - mark as AUC_{0-t} considering τ as a t , given we do not know τ in this exercise. E.g.

PK 1

A lower clearance of levofloxacin (<2 mL/min/kg), conditioning both a longer elimination half-life (approximately 9 h) and a larger $AUC(0-\tau)$ (approximately 80 mg/L $\times$ h), was observed for both routes in our patients than in healthy volunteers.

comments

Tau is not exactly t but I assumed this would be acceptable.

- Relative Bioavailability relative from one condition to another instead of F_{rel} - Please label this as NIL (a covariate effect and out of scope).

PK 1

Mean relative bioavailabilities calculated from these data were 83% and 86% respectively.

comments

I think this is tricky, it is relative from one condition to another. This is different compared to relative F in a popPK model.